Jeffrey Costello MS Mechanical Engineering | Highly Interdisciplinary, Perseverant

Portfolio: jdcostllo.github.io **jdcostllo@gmail.com** **Linkedin** **Somerville, MA**

Education

MS Mechanical Engineering 2022 - September 2024 (GPA 5.0 / 5.0)
Massachusetts Institute of Technology (MIT) Cambridge MA

BS Mechanical Engineering 2013 - May 2017 (GPA 3.41 / 4.00)
Boston University Boston, MA

Professional Experience, 9 Years

Eden GeoPower

Senior Mechanical Engineer, Feb 2026 - Present

Own mechanical design, procurement, and assembly of 400 kV pulsed-power equipment. Direct on-site field operations as needed to ensure the successful deployment of pulsed-power systems, surface instrumentation, and downhole tools.

Irradiant Technologies

Engineering Team Lead, Feb 2025 - Feb 2026

Managed a 6-person engineering area, driving cross-subteam collaboration and IP aggregation. Led SOP creation, conducted biannual performance reviews, and aligned area initiatives with senior leadership.

Senior Mechanical Engineer, Oct 2024 - Feb 2026

Designed automated dual-axis tilt compensation mechanisms for lithography machines and novel optics/materials solutions. Implemented company-wide Functional Requirements protocols and managed engineering and nanotechnology interns. Led in-house prototyping using CNC equipment.

MIT Global Engineering and Research (GEAR) Center

Research Assistant (Masters Student) Sep 2022 - Sep 2024

Staff Engineer Oct 2021 - Sept 2022 and Jan 2018 - Sept 2019

Developed parametric modeling and genetic algorithm optimization for solar-powered desalination (TEDR) systems. Built experimental hardware, test stands, and software for field pilots, managing international collaborations and in-field validation.

BU Engineering Product Innovation Center

Lab Supervisor / Manufacturing Instructor Sep 2019 - Oct 2021

Managed daily operations, safety, and throughput for a 15,000 sq ft machine shop serving 1,000+ students per semester. Developed and directed laboratory exercises for the Automated Design and Manufacturing Laboratory.

Chant Engineering Company, Inc

Mechanical Engineer Jun 2017 - Dec 2017

Led multiple concurrent engineering projects from conception through fabrication and delivery. Executed core mechanical design, drafting, advanced calculations, and ISO 9001 documentation.

Objective

Technical leader with extensive experience directing engineering teams and managing cross-functional startup initiatives. Looking to leverage a strong background in precision machine design, hardware-software integration, and team mentorship to solve complex, interdisciplinary systems engineering problems.

Skills

Programming, Automation
Python, MATLAB, Linux, Ladder (PLC), HTML/CSS/Javascript, RSLogix, Universal Robots, Teledyne-Dalsa Vision Systems

System Integration, Electronics
CAN, MQTT, RS232/RS485 Serial, Modbus, Relay Logic, Power Electronics

CAD/CAE/CAM
Solidworks, Onshape, PTC Creo, Fusion, Solidworks FEA, GibbsCAM, HSM Express

Manufacturing
Wire EDM, Manual/CNC Mill and Lathe, FDM and SLA 3D Printers, Laser Cutter, Waterjet

Project Management, Quality Assurance
ISO 9001, Functional Requirements, Technical Documentation, Teaching

Notable Coursework (MS 2024)

MIT 6.2222 Power Electronics Laboratory (Fall 2023)
Designed and built a custom "Camera-Slide" featuring scratch-designed circuitry, a switched-capacitor power supply, a boost converter, a custom stepper motor driver, and an Infineon PSoC.

MIT 2.720 Elements of Machine Design (Spring 2023)
Modeled, designed, and fabricated a competition-winning precision desktop lathe utilizing homogeneous transformation matrices and precision machine design principles.